

Physical Science Part 5

1. Which statement about the North-magnetic pole of the Earth is TRUE?
 - A. Its location never changes.
 - B. It corresponds to the N-pole of a bar magnet.
 - C. It corresponds to the S-pole of a bar magnet.
 - D. It is at the same location as the geographic north pole of the Earth

2. An iron nail can be turned into a temporary magnet if it is held in a strong magnet field. Which method of magnetization is this?
 - A. Charging
 - B. Conduction
 - C. Introduction
 - D. Saturation

3. Which of the following factors affect the size of the induced current in a coil?
 - I. Number of turns in the coil
 - II. Size of the inducing field
 - III. Direction of the winding in the coil
 - IV. Resistance of the wire in the coil
 - A. I and II only
 - B. III and IV only
 - C. I, II, and III only
 - D. I, II, and IV only

4. To use the right hand rule to predict the direction of an induced current in a coil, your thumb must point _____.
 - A. to the direction of the motion of the wire
 - B. to the S-pole of the induced magnetic field
 - C. in the direction of the induced current
 - D. in the direction of the inducing field

5. When a bar magnet is inserted into a coil, a current of 12.0 A is induced in the coil. If four magnets of the same type are inserted into a coil at the same time, what would be the induced current?
 - A. 3 A
 - B. 4 A
 - C. 12 A
 - D. 48 A

6. Which of the following involves adsorption?
 - A. Plating the copper on a steel object
 - B. Adherence of paint to wood surfaces
 - C. Brown color of the eyes of the Filipinos
 - D. Removal of odor inside a refrigerator using charcoal

7. Which of the following BEST describes temperature?
- A. Temperature measures the total energy in something.
 - B. Temperature measures the kinetic energy in substance.
 - C. Temperature measures the total kinetic energy contained in an object.
 - D. Temperature measures the average molecular kinetic energy in substance.
8. What is the direction of a magnetic field within a magnet?
- A. From north to south
 - B. From south to north
 - C. Front to back
 - D. Back to front
9. Which is the induced voltage across a stationary conductor in a stationary magnetic field?
- A. Decreased
 - B. Increased
 - C. Zero
 - D. Reversed in polarity
10. What is the magneto motive force (mmf) of a wire with 8 turns carrying 3 A of current?
- A. 2.67
 - B. 24
 - C. 240
 - D. 2400
11. A coil of wire is placed in a changing magnet field. If the number of turns in the coil is decreased, what will happen to the voltage induced across the coil?
- A. Remain constant
 - B. Be excessive
 - C. Decrease
 - D. Increase
12. Which of the following is TRUE about the frequency and velocity of electromagnetic waves?
- A. Frequency and wavelength are interdependent of each other
 - B. As frequency increases, wavelength decreases
 - C. As frequency increases, wavelength increases
 - D. Frequency is constant for all wavelengths
13. Which of the following correctly lists electromagnetic waves in order from longest to shortest wave length?
- A. Television, infrared, visible light, X-rays
 - B. Gamma rays, ultraviolet, infrared, microwaves
 - C. Radio waves, infrared, gamma rays, ultraviolet
 - D. Micro waves, ultraviolet, visible light, gamma rays

14. An electromagnetic wave is radiated by a straight wire antenna that is oriented vertically. What should be the orientation of a straight wire receiving antenna?
- A. Diagonally
 - B. Vertically
 - C. Horizontally and in a direction parallel to the wave's direction of motion
 - D. Horizontally and in a direction perpendicular to the wave's direction of motion
15. What should a good diode tested with an ohmmeter indicate?
- A. High resistance when forward or reverse biased
 - B. Low resistance when forward or reverse biased
 - C. High resistance when reverse biased and low resistance when forward biased
 - D. High resistance when forward biased and low resistance when forward biased
16. Vacuum tubes in a radio transmitter are used to _____.
- A. record programs
 - B. provide DC supply
 - C. generate high power radio waves
 - D. provide lighting inside the studio
17. Which material is the usual component of an anode?
- A. Aluminum
 - B. Carbon
 - C. Copper
 - D. Nickel
18. A metal spoon is placed in a cup filled with hot coffee. After sometime, the exposed end of the spoon becomes hot even without a direct contact with the liquid. What explains this phenomenon?
- A. Conduction
 - B. Convection
 - C. Radiation
 - D. Thermal expansion
19. Which cause the warmth that you fell when you place your finger at the side of the flame of a burning candle?
- A. Conduction
 - B. Convection
 - C. Radiation
 - D. Not enough information given
20. Which process of heat transfer happens by the movement of mass from one place to another?
- A. Conduction
 - B. Convection
 - C. Induction
 - D. Radiation

21. When work is done by a system, what happens to its temperature assuming that no heat is added to it?

- A. Decreases
- B. Increases
- C. Remains the same
- D. None of these

22. Which states that “when two systems are in thermal equilibrium with a third system, then they are in thermal equilibrium with each other?”

- A. Zeroth Law of Thermodynamics
- B. First Law of Thermodynamics
- C. Second Law of Thermodynamics
- D. Mechanical Equivalent of Heat

23. What happens in an adiabatic process?

- A. No heat enters or leaves the system.
- B. The pressure of the system remains constant.
- C. The temperature of the system remains constant.
- D. The system does no work nor work is done on it.

24. What thermodynamic process is involved in a closed car inside a hot garage?

- A. Adiabatic
- B. Isobaric
- C. Isochoric
- D. Isothermal

25. Consider the thermal energy transfer during a chemical process. When heat is transferred to the system, the process is said to be _____ and the sign of H is _____.

- A. endothermic, positive
- B. endothermic, negative
- C. exothermic, positive
- D. exothermic, negative

26. Which of the following statements is FALSE?

- A. Entropy increases with the number of microstates of the system.
- B. Any irreversible process results in an overall increase in entropy.
- C. The total entropy of the universe in any spontaneous process.
- D. The change in entropy in a system depends on the initial and final states of the system and the path taken from one state to the other.

27. During a refrigeration cycle, heat is rejected by the refrigerant in _____.

- A. compressor
- B. condenser
- C. evaporator

D. expansion valve

28. Which of the following CORRECTLY describes a refrigeration system?

- A. Rejects energy to a low temperature body.
- B. Rejects energy to a high temperature body.
- C. Removes heat from a low temperature body and delivers it to a high temperature body.
- D. Removes heat from a high temperature body and delivers it to a low temperature body.

29. Which of the following refrigerant has the maximum ozone depletion potential in the stratosphere?

- A. Ammonia
- B. Carbon dioxide
- C. Fluorine
- D. Sulfur dioxide

30. Which phenomenon is BEST explained by the particle nature of light?

- A. Doppler effect
- B. Interference
- C. Photoelectric effect
- D. Polarization

31. According to the quantum theory of light, the energy of light is carried in discrete units called ____.

- A. alpha particles
- B. photoelectrons
- C. photons
- D. protons

32. Rutherford observed that most of the alpha particles directed at a metallic foil appear to pass through unhindered, with only a few deflected at large angles. What did he conclude?

- A. Atoms can easily absorb and re-emit alpha particles
- B. Atoms consist mainly of empty space and have small, dense nuclei.
- C. Alpha particles behave like waves when they interact with atoms.
- D. Atoms have most of their mass distributed loosely in an electron cloud.

33. When an electron changes from a higher energy to a lower energy state within an atom, a quantum of energy is _____.

- A. absorbed
- B. emitted
- C. fissioned
- D. fused

34. When does nuclear fission occur?

- A. When we cut nuclei into two with a very small cutting device
- B. When one nucleus bumps into another causing a chain reaction

- C. When a nucleus divides spontaneously, with no apparent reason
- D. When electrical force inside a nucleus overpower forces

35. When does nuclear fusion release energy?

- A. When uranium emits a neutron
- B. When heavy ions fuse together
- C. When very light nuclei fuse together
- D. When uranium splits into two fragments

36. Albert Einstein's Second postulate of Special Relativity state that the speed of light_____.

- A. can decrease if the speed of the observer decreases
- B. can increase if the speed of the observer increases
- C. randomly changes depending upon its original light source
- D. is constant regardless of the speed of the observer or the light source

37. A passenger train travels east at high speed. One passenger is located at the east side of one car, another is located at the west side of that car. In the train's frame, these two passengers glance up the same time. In the earth's frame _____.

- A. they glance up simultaneously
- B. the passengers glance sideways
- C. the passenger at the east side glances up first
- D. the passenger at the west side glances up first

38. Benie11 travels at high speed from Earth to the star Alpha Centauri which four light years away. Which one is CORRECT in Benie11's frame?

- A. The clocks on earth and on Alpha Centauri are synchronized.
- B. The trip takes more time than it does in the earth's frame
- C. Benie11 travels to Alpha Centauri over a length that is shorter than four light years.
- D. Alpha Centauri travels to Benie11 over a length that is shorter than four light years.

39. One clock is placed on top of a mountain and another clock is at the bottom the mountain. Ignore the rotation of the earth. Which clock ticks more slowly?

- A. They tick at the same rate.
- B. The one on top of the mountain.
- C. The one at the bottom of the mountain.
- D. Cannot be determined.

40. You are in a windowless spacecraft. What do you need to do determine whether your space ship is moving at constant non-zero velocity, or is at rest, in an inertial frame of earth?

- A. Make very precise time measurements.
- B. Make very precise mass measurements.
- C. Make very precise length and time measurements.
- D. Nothing, because you cannot succeed no matter what you do.

Answer Keys:

1. C
2. C
3. D
4. A
5. D
6. D
7. D
8. A
9. C
10. B
11. C
12. B
13. A
14. B
15. C
16. C
17. A
18. A
19. C
20. B
21. A
22. A
23. A
24. C
25. A
26. D
27. B
28. D
29. C
30. C
31. C
32. B
33. B
34. D
35. C
36. D
37. D
38. D
39. C
40. D

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